



HK IMO Training 2022-2023
Problem Set 9
For 3 December 2022



Problem 33. Let a, b, c be given real numbers where $a \neq 0$. Prove that there is a polynomial $P(x)$ with real coefficients such that $x^2 + 1$ is a factor of $aP(x)^2 + bP(x) + c$.

Problem 34. Find all integers a, b, c, n such that $\frac{(a-b)(b-c)(c-a)}{2} + 2 = 2016^n$.

Problem 35. In acute $\triangle ABC$, $AB \neq AC$. Let Γ be the circumcircle of $\triangle ABC$, and let AA' be a diameter of Γ . Let D be the point on Γ such that $AD \perp BC$. A point E is chosen such that $BDCE$ forms a parallelogram. Let M be the midpoint of BC . A point P is chosen on the same side of BC as A such that $\angle BPM = \angle ACB$ and $\angle CPM = \angle ABC$. Let Q be the second intersection point of Γ and the circumcircle of $\triangle A'EP$. Prove that A, M, Q are collinear.

Problem 36. Let k be a positive integer and let G be a simple directed graph. Suppose each vertex has in-degree at least $3k + 1$ or has out-degree at least $3k + 1$ (or both). Prove that we can partition G into two nonempty subgraphs such that each vertex in each subgraph has in-degree at least k or has out-degree at least k .